

Predictive Maintenance for Robotic Arms

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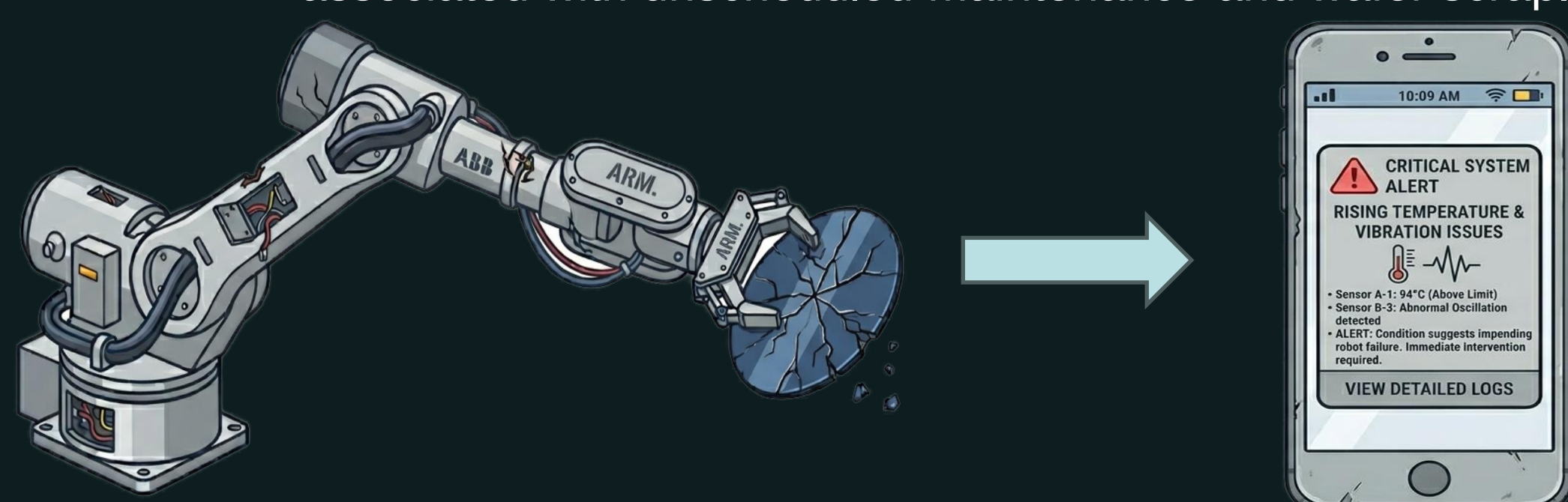


Abstract

Develop an AI system that predicts robot arm failures in motors, joints, or sensors of the robotic arm using sensor data such as vibration, current, or temperature

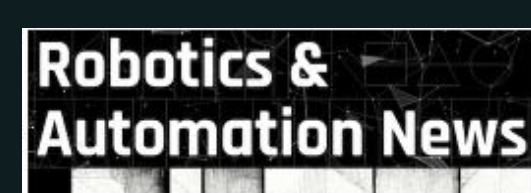
Objective

The objective is to integrate multi-modal sensor **telemetry** (temperature, current, and vibration) into a predictive AI model. By identifying early-stage anomalies, the system alerts stakeholders to impending failures, effectively reducing the financial **overhead** associated with unscheduled maintenance and wafer scrap.

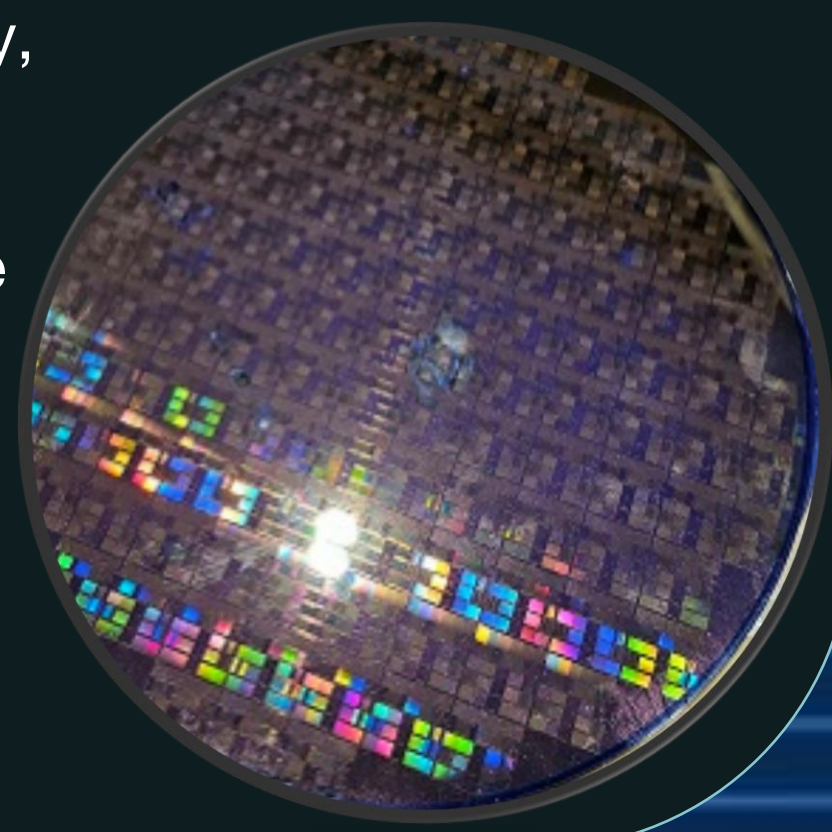


Background Research

Warehouses and factories make use of robotic arms for continuous operation within their network. Errors from several factors may occur such as issues with a robot arm's joint or motors. Since there is a demand for programs that can help minimize downtime, companies like IBM have developed products such as the Maximo Application Suite which try to help with AI and machine learning to allow for a dynamic schedule by performing predictive maintenance.



- Semiconductor manufacturing encompasses photolithography, etching, deposition, and metrology
- Current maintenance strategies based on time intervals fall short in addressing unexpected failures and recovery in ample time



Research Findings



Temperature – It is normal for temperatures to gradually increase with a robot's continuous use, but when thresholds are met, stakeholders should be notified



Vibration – Vibration notifications can be sent out when there are sharp vertical spikes which indicate sudden mechanical shocks or misalignments



Current – When the current spikes from around the mean value, we can send notification that the motor is working harder to overcome mechanical resistance

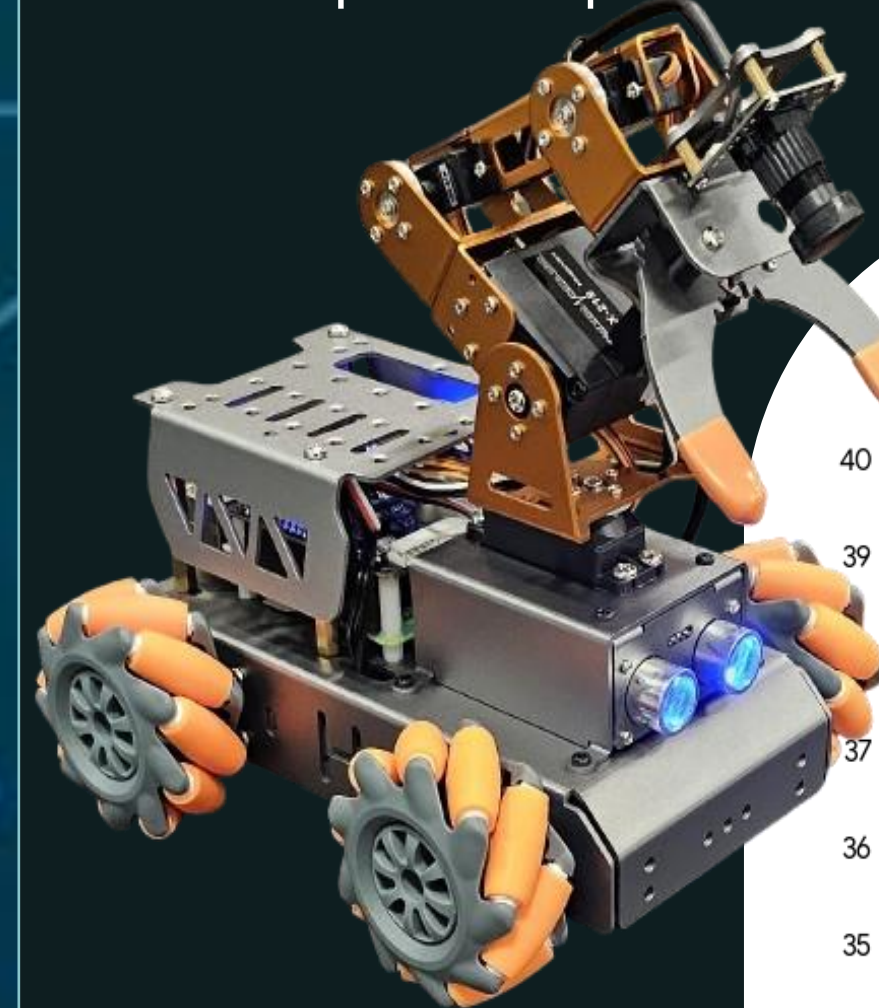
Arm enables predictive fault detection through dual use of their Cortex-A and Cortex-M processors. Cortex-A uses AI to reason about 'sensor fusion' (combining vibration, temperature, and current) data collected by Cortex-M and make precise decisions. The complex computations happen right on the edge – reducing latency and room for error caused by cloud disruptions. Having a reasoning system and a reflex system working simultaneously in tandem with robot operating system (ROS) is one way to predict errors before they occur in robotic arms.



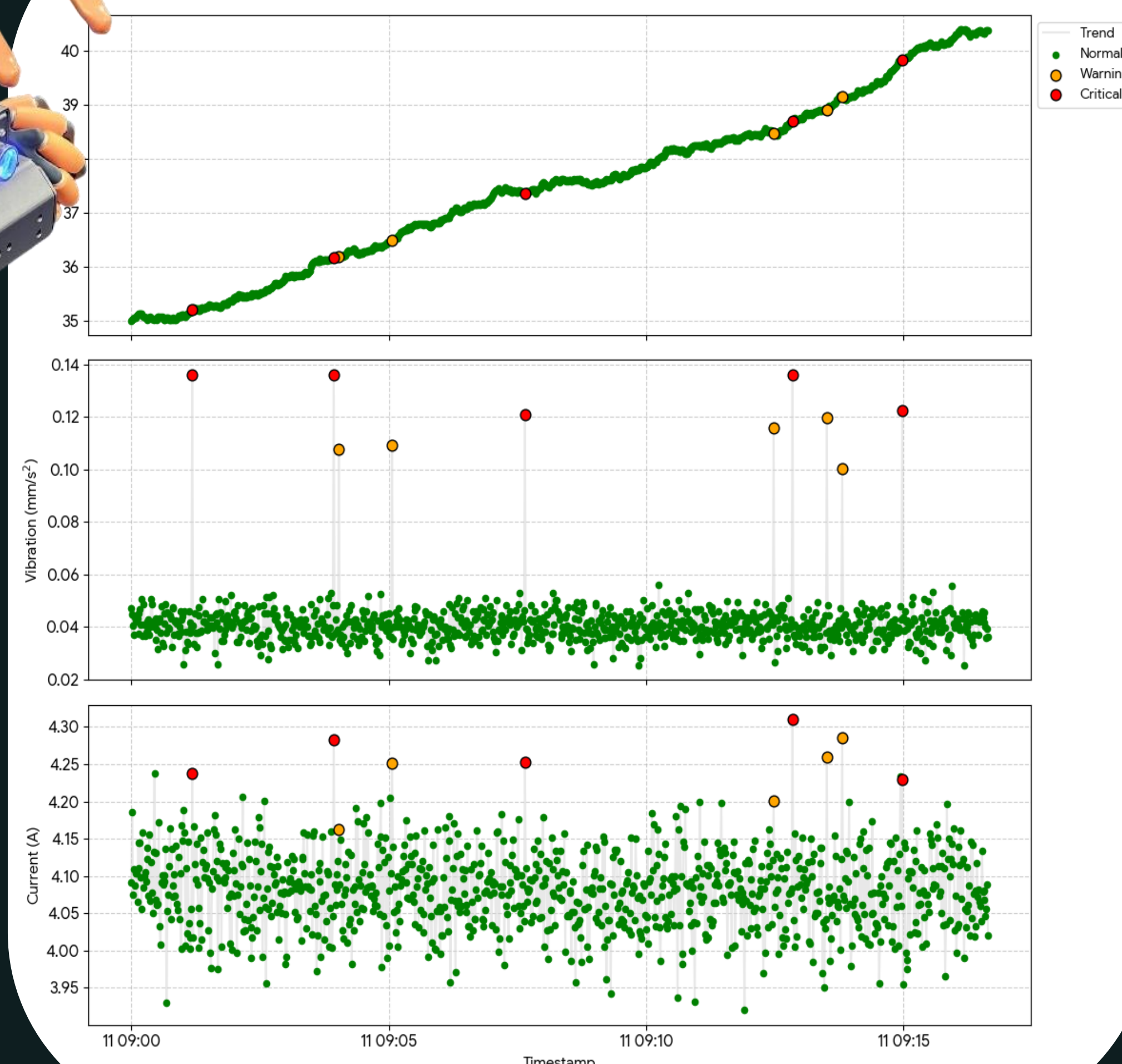
Results

Below you can see the timeseries graph of a robotic arm's sensor data for temperature, vibration, and current.

- **Normal (Green):** Parameters are within safe operating limits.
- **Warning (Orange):** Parameters have exceeded the first threshold; maintenance should be scheduled.
- **Critical (Red):** Parameters have exceeded the safety limit; immediate intervention is required to prevent hardware failure.



Robot Sensor Data Monitoring (Temperature, Vibration, Current)



Future Work

Beyond the semiconductor sector, this predictive maintenance framework is designed for seamless integration into any automated manufacturing environment. High-precision industries—including automotive assembly, aerospace engineering, and pharmaceutical packaging—can leverage these AI models to preemptively identify mechanical fatigue. By transitioning from reactive to proactive maintenance, manufacturers can eliminate costly production bottlenecks, ensure consistent output quality, and significantly maximize the operational lifecycle of robotic assets.



Citations

Tao, M. (2025, August 15). Top 10 predictive maintenance companies using AI to keep robotic systems running smoothly. Robotics and Automation News. <https://roboticsandautomationnews.com/2025/08/15/top-10-predictive-maintenance-companies-using-ai-to-keep-robotic-systems-running-smoothly/93602/>

R. C. Palsaniya, "Predictive Maintenance in Semiconductor Manufacturing - AI Machine Learning Application for Downtime Reduction," 2025 IEEE 34th International Symposium on Industrial Electronics (ISIE), Toronto, ON, Canada, 2025, pp. 1-5, doi: 10.1109/ISIE62713.2025.11124753. <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=11124753&isnumber=11124555>

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Support Information

